



### Question 1

(a) Fig 1.1 below shows a section of a cell.

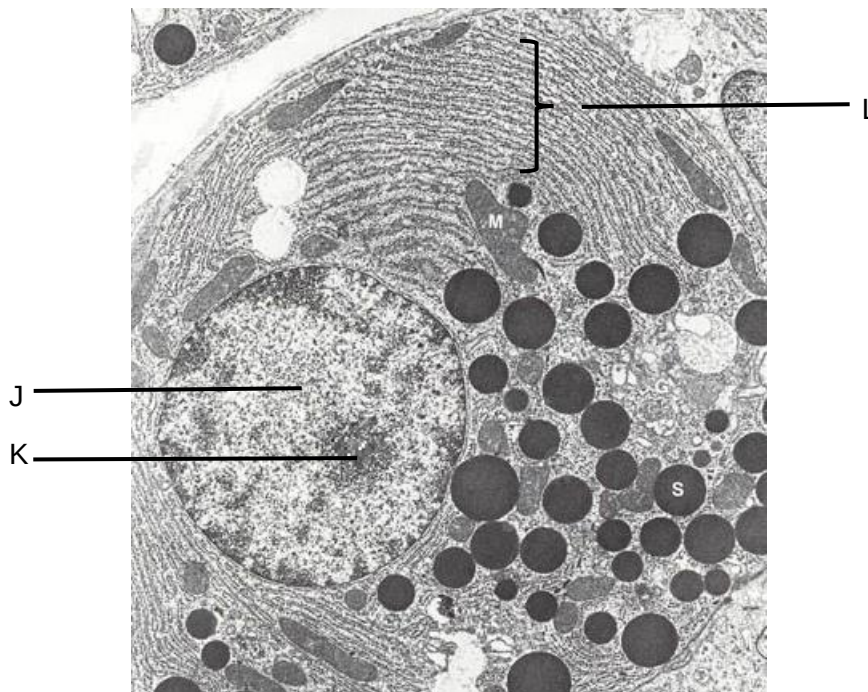


Fig. 1.1

(<https://www.visualhistology.com/cells/cells/>)

(a) Organelles J and K differ in their sizes.

Describe one other difference in the structure between J and K. [2]

- J is bounded by a double membrane known as the nuclear envelope whereas
- K is not bounded by any membrane;

(b) Both DNA and RNA are biomolecules that contain genetic information, they perform different functions.

(i) Using your knowledge of DNA structure, explain why DNA is structurally more adapted to be a store of genetic information compared to the RNA. [3]

- DNA is **double stranded** whereas RNA is single stranded:
  - DNA more stable as the hydrophobic bases are projecting into the helix, protected from exposure to mutagens;

- DNA undergoes semiconservative replication- each strand can be used as a template for the synthesis of its complementary strand;
- allows accurate copying of the genetic information to be inherited by the next generation;  
Or if there is an error/mutation in one of the strand, the other intact strand can be used to guide repair/ error correction;
- DNA is **more stable**: genetic information can be stored without changes;
  - as it contains deoxyribose sugars which are less reactive than ribose sugars;
  - double stranded with large number of hydrogen bonds between nitrogenous base pairs across the two strands of polynucleotide;
  - **Hydrophobic interactions** between the stacked bases;
  - Lower reactivity as there are **no hydroxyl groups** at carbon 2 of deoxyribose;
- AVP;

(ii) Different types of RNA can be found associated with L.

Briefly describe the functions of these RNA. [3]

- Mature mRNA that encodes the base sequence for the protein to be synthesised from the nucleus to the cytoplasm for translation;
- Ribosomal RNA/ rRNA – together with ribosomal proteins, form the ribosomes which are sites of protein synthesis/translation;  
Or allows ribosome to binds to mRNA for translation;
- Ribozyme/peptidyl transferase that catalyses the formation of peptide bonds between amino acids during translation;
- Transfer RNA/ tRNA (adaptor molecule) that carries the correct amino acid to the ribosome for translation based on anticodon-codon binding;

(c) Biomolecules such as enzymes perform important functions.

Gastric lipase consists of 379 amino acid residues. It is active in the highly acidic stomach environment. The active site consists of three important amino acids – serine, histidine, aspartic acid at positions 153, 353 and 324 respectively.

(i) Explain how the three amino acids are brought together at the active site. [3]

- Tertiary structure - extensive folding and coiling of the polypeptide upon itself to form a specific 3D shape/conformation;
- Due to interactions between R groups of amino acids that are far away;
- Type of interactions: hydrogen, ionic, hydrophobic interactions and disulfide;
- Resulting in the 3 amino acids being placed close to each other;

(ii) Suggest how gastric lipase is able to function at a highly acidic medium. [2]

- Higher proportion of covalent bond such as disulfide linkages;

- Lower proportion of Hydrogen bonds and ionic interactions;
- able to retain the 3D conformation so that the shape of the active site is maintained;

**(d)** Metabolic reactions in cells occur in steps.

Describe two advantages of metabolic reactions occurring in steps. [2]

[Any 2, examples are only for understanding]

(A) Controlled release of energy

- the energy can be efficiently captured to form ATP;
  - heat released can also be controlled so that enzymes do not get denatured;
- [Example – respiration where gradual release of energy through the stages of glycolysis, link reaction, Krebs's cycle and oxidative phosphorylation]

(B) Control & regulation

- Allows control and regulation at each step of the pathway so that there is no overproduction of products;
- Intermediates from steps can be channelled / utilized as reactants in the subsequent steps of the reaction;
- Example: glycolysis
- Enabling negative feedback OR end-product inhibition;
  - Without feedback inhibition, energy or raw materials that could be used for important cellular functions might be wasted on unnecessary ones;
  - Prevents depletion - Without feedback inhibition, raw materials and energy might be depleted by biochemical processes that don't stop, even when their end product is not no longer needed;
- Provides alternate pathways if there is a block in one of the step - eg. from aerobic to anaerobic pathway when oxygen becomes unavailable;

[total: 15]

## Question 2

Type 1 diabetes is a chronic disease. In people with type 1 diabetes, cells in the pancreas that make insulin are destroyed, and the body is unable to make insulin. Scientists investigated the use of induced pluripotent stem cells (iPS cells) to treat type I diabetes in mice.

During the experiment, they stimulated the *in vitro* differentiation of iPS cells into pancreatic cells. The scientists then set up three experimental groups:

- Group **A** – 30 mice with type I diabetes received pancreatic cell transplants derived from iPS cells.
- Group **B** – 30 mice with type I diabetes were left untreated.
- Group **C** – 30 mice without diabetes were left untreated.

The scientists measured the blood glucose concentration of all the mice on a weekly basis for 12 weeks. The results obtained are shown in Fig. 2.1.

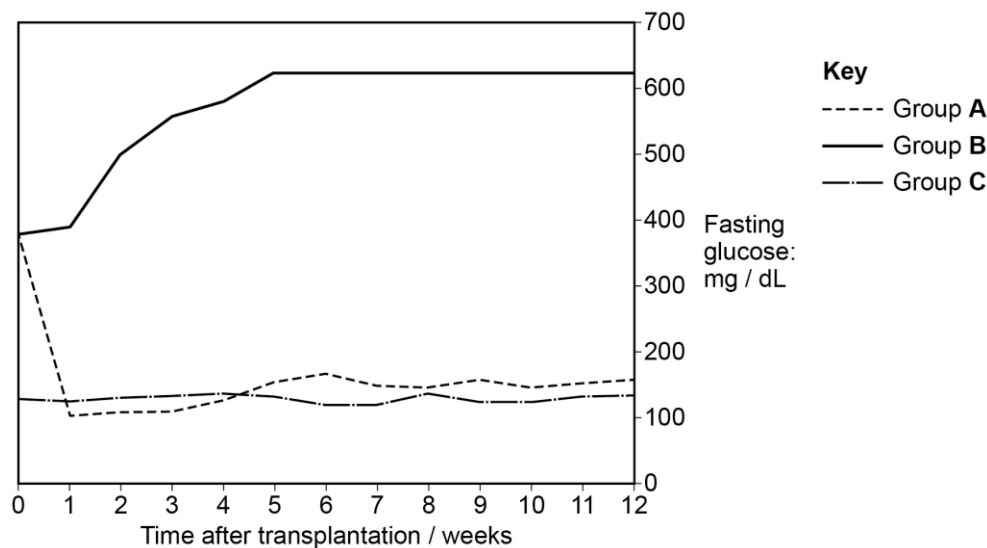


Fig. 2.1

- (a) With reference to Fig. 2, comment on the effectiveness of the iPS treatment. [2]
- (i)
- Effective because fasting glucose levels was much lower for group A mice compared to group B mice without the treatment (QV: 160mg/dl vs 620mg/dl);
  - Effective because fasting glucose levels for group A mice at similar levels compared to group C mice (160mg/dl to 150mg/dl);
- (ii) Explain why iPS was used in this treatment instead of hematopoietic stem cells? [2]
- iPS being pluripotent vs hematopoietic stem cells being multipotent;
  - idea that iPS can be programmed to differentiate into almost any type of cells such as pancreatic cells vs hematopoietic stem cells only able to differentiate into different blood cells / limited range of cell types;

- (iii) Suggest why the study on iPS cell treatment for type 1 diabetes was conducted on mice. [1]
- idea of ethical considerations – unknown health risks / hazards to humans if iPS cells are introduced as a new treatment;
  - Similar hormones / metabolic or cellular processes in human and mice;

- (b) A scientist stained and photographed all the chromosomes present in one of the iPS cell during mitosis.



Fig. 2.2

- (i) Name the stage of mitosis shown in Fig. 2.2. [1]
- Prophase;
- (ii) Explain two pieces of observable evidence from Fig. 2.2 that the iPS cell was undergoing mitosis. [2]
1. Each chromosome is made up of two chromatids because DNA has been replicated;
  2. DNA / chromatin has undergone condensation to form chromosomes which occurs during mitosis;
  3. The homologous chromosomes did not pair up / did not form bivalents / did not form chiasmata which would normally occur during meiosis;
- Max 2

- (iii) Explain the significance of mitosis for the iPS cells treatment of type 1 diabetes in the group A mice. [2]
- Mitosis create genetically identical pancreatic cells which can secrete insulin in the mice;
  - Mitosis can create large number of pancreatic cells to replace and replenish the destroyed pancreatic cells in mice;

- (c) Studies conducted on the group A mice revealed traces of cancerous cells in the skin two years after the treatment. It was postulated that the cancerous cells arise as a result of the iPS cell treatment.
- (i) Suggest why iPS cells might potentially become cancerous. [3]
- iPS cells can undergo long term self-renewal;
  - idea of accumulation of genetic mutations in the cells after many rounds of cell division;
  - leading to cells dividing excessively and without control (idea of loss of control in cell cycle);
- (ii) Explain the significance of events that must occur in order for cancerous cells to be located at the different organs. [2]
- metastasis – cancerous cells invade normal tissue / blood vessels / lymph vessels in order to migrate to other parts of the body;
  - angiogenesis – formation of new blood vessels to supply nutrients and oxygen to the dividing cancerous cells in the new sites;

[total: 15]

### Question 3

Table 2.1 compares the mean milk yield of the first milk-producing period (first lactation) and the next milk-producing period (second lactation) for Xinong Saanen goats of each genotype.

Table 2.1

| genotype at position 5752 | mean milk yield / kg |                  |
|---------------------------|----------------------|------------------|
|                           | first lactation      | second lactation |
| CC                        | 679                  | 940              |
| CG                        | 622                  | 858              |
| GG                        | 616                  | 834              |

- (a) (i) Variation in a phenotypic characteristic such as milk yield is caused by a combination of genetic and environmental factors.

Goats also show variation in milk yield between the first lactation and second lactation.

Suggest, with reasons, whether the variation in milk yield between the first lactation and second lactation, as shown in Table 2.1, is genetic or environmental. [3]

1. environmental / not genetic;
2. (mean) milk yield increases from 1st to 2nd lactation in all genotypes / increase for all genotypes are about the same (if calculate percentage increase);
3. [Data to support] mean milk yield increase from 679 to 940 kg for CC genotype, 622 to 858 kg for CG genotype and 616 to 834 kg for GG genotype / increase by 218 - 261 kg for all genotypes
4. [Any idea of] older / 2nd lactation, goats have, bigger / more developed, udders/ age / maturity;

- (ii) The variation at position 5752 of the gene coding for a growth factor is due to a substitution mutation from **G** to **C**.

With reference to Table 2.1, describe the significance of the substitution from **G** to **C**. [2]

1. (substitution / **G** → **C** / **CC**) increases milk, yield / production;
2. little difference between **GG** and **CG** / substitution from **GG** to **CG** not as significant increase as compared to substitution from **CG** to **CC**;
3. [Data to support] In first lactation, between GG and CG, slight increase from 616 to 622 kg/ only 6 kg increase compared to between CG and CC, significant increase from 622 to 679 kg/ greater increase of 57 kg;

- (b) A breeder is interested in the inheritance of colour coat in goats. Research showed that it is determined by a series of multiple alleles. Three alleles each producing a different coat colour have been identified. Fig. 2.1 shows the inheritance of coat colour in one family, where generation I is a cross of Guanzhong male with Xinong Saanen female.



Parental Genotypes:

$C^G C^B$

x

$C^W C^B$

meiosis  
↓

meiosis  
↓

Gametes:

$C^G$

$C^B$

$C^W$

$C^B$

Fertilisation (Punnet Square):

|  |       |                         |                         |
|--|-------|-------------------------|-------------------------|
|  |       | $C^W$                   | $C^B$                   |
|  | $C^G$ | $C^W C^G$<br>White coat | $C^G C^B$<br>Grey coat  |
|  | $C^B$ | $C^W C^B$<br>White coat | $C^B C^B$<br>Black coat |

F2 phenotypic ratio: 2 white coat : 1 grey coat : 1 black coat goats

III-5

III-3

III-4

1. Correct symbols used
2. Correct parental (phenotypes &) genotypes
3. Correct gametes
4. Correct F2 phenotypic ratio shown / relate genotype to phenotype of generation III-3, 4 and 5;

[Total : 15]

4 (a) Outline briefly that photosynthesis involves conversion of light energy to chemical energy and this energy is stored in the form of carbohydrates. [5]

1. Photosynthesis can be divided into light dependent reaction and light independent reaction / Calvin cycle
2. Light dependent reaction occurs at the chloroplast thylakoid
3. Light energy is converted into chemical energy as ATP and reduced nicotinamide adenine dinucleotide phosphate (reduced NADP)
4. Water undergoes photolysis to give rise to oxygen as a by-product
5. Photochemical products ATP and reduced NADP produced in light dependent reaction will be used in light independent reaction / Calvin cycle
6. Light independent reaction / Calvin cycle occurs in the chloroplast stroma
7. ATP and reduced NADP are used to power the chemical reduction of CO<sub>2</sub>
8. Final product is glucose/ sugar which complexed to form starch for storage

(b) Compare the structure and properties of cellulose and starch and relate these to their functions. [10]

|                                   | Cellulose                                                                                                                                                                                                                                        | Starch                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monomer (1m)                      | <ul style="list-style-type: none"> <li>Consists of <math>\beta</math> glucose as monomers held together</li> </ul>                                                                                                                               | <ul style="list-style-type: none"> <li>Consists of <math>\alpha</math> glucose as monomers held together</li> </ul>                                                                                                                                                                                                                                                |
| Bond (1m)                         | <ul style="list-style-type: none"> <li><math>\beta(1, 4)</math> glycosidic bonds</li> </ul>                                                                                                                                                      | <ul style="list-style-type: none"> <li><math>\alpha(1, 4)</math> &amp; <math>\alpha(1, 6)</math> glycosidic bonds.</li> </ul>                                                                                                                                                                                                                                      |
| Shape (2m)                        | <ul style="list-style-type: none"> <li>Alternate glucose residues have to be rotated <math>180^\circ</math> thus resulting in a <u>long and straight chain</u> in the polymer</li> </ul>                                                         | <ul style="list-style-type: none"> <li>Polymer twisted into a <u>helical</u> shape for amylose &amp; <u>Branching</u> for amylopectin</li> </ul>                                                                                                                                                                                                                   |
|                                   |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                    |
| Property (1m)                     | Insoluble in water                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                    |
| Explanation (max 4m)              | <ul style="list-style-type: none"> <li>straight chain is that most of the hydroxyl groups of glucose residues are projected in all directions from each chain.</li> <li>Hydroxyl groups form hydrogen bonds with neighbouring chains.</li> </ul> | <ul style="list-style-type: none"> <li>Most of the hydroxyl groups of glucose residues are projected into the interior of the helices.</li> <li>Hydroxyl groups form hydrogen bonds with each other to stabilize the shape and also make it insoluble in water.</li> <li>Can function as storage molecule without affecting the water potential of cell</li> </ul> |
|                                   |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                    |
| Function (1m)                     | <ul style="list-style-type: none"> <li>structural function – allows maintenance of integrity of the cell wall</li> </ul>                                                                                                                         | <ul style="list-style-type: none"> <li>large store of carbon (building block) and energy</li> </ul>                                                                                                                                                                                                                                                                |
|                                   |                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                    |
| Relate structure to function (3m) | <ul style="list-style-type: none"> <li>cross-linking binds the chains rigidly together conferring great tensile strength and inelastic nature</li> </ul>                                                                                         | <ul style="list-style-type: none"> <li>structure is compact and ideal for storage as many glucose residues can be stored in a small volume within the cell.</li> <li>Many branch points / highly branched – idea of many access points for enzyme to break down</li> </ul>                                                                                         |

|  |  |                                               |
|--|--|-----------------------------------------------|
|  |  | molecule leading to release of energy readily |
|  |  |                                               |

QWC: Answers must be presented by clearly comparing cellulose to starch.

5 (a) Explain the importance of meiosis and random fertilization in evolution. [5]

- Meiosis and random fertilisation generates genetic variation in a population via reshuffling of existing alleles;
- Meiosis
  - results in production of haploid gametes that are genetically variable;
  - via crossing over between non-sister chromatids of homologous chromosomes and independent assortment of homologous chromosomes;
- Random fertilisation
  - Random fusion of genetically variable haploid gametes results in new combinations of alleles and genotypes in individuals;
- Variation is the raw material for natural selection;
  - Genetic variation must be expressed in the phenotypes for natural selection to take place;
  - Individuals with favourable phenotypes will have higher reproductive success ( ie. will be selected for, survive longer, reproduce and pass on their favourable alleles to their offspring) while those without the favourable phenotypes will have lower reproductive success;
  - Allele frequencies of the favourable alleles (coding for the favourable phenotypes) will increase in subsequent generation;

(b) Explain how genetic variation, including harmful recessive alleles, may be preserved in a natural population. [10]

- Variation (phenotypic) refers to the **differences in characteristics between individual organisms** belonging to the same natural population;
- Mechanisms to preserve variation
  - Heterozygote protection
    - In diploid organisms, the heterozygotes showing the dominant phenotype carries the recessive allele which is masked by the dominant allele/ not expressed in the phenotype;
    - recessive allele will may persists by propagation of the heterozygotes in the population;
  - Balancing selection
    - Balancing selection occurs when natural selection maintains two or more alleles in the gene pool of the population;
    - Heterozygote advantage  
Heterozygotes have greater fitness over both kinds of homozygotes and thus both alleles are maintained;
    - Use sickle cell anaemia to illustrate the 'harmful' recessive allele that persists in malarial regions;
    - Frequency-dependent selection
    - In frequency-dependent selection, the fitness of a particular phenotype declines if it becomes too common in the population;
    - Frequencies of the different phenotypes oscillate over time and are maintained within the gene pool;
  - Neutral variation
    - Genetic variations that have little or no impact on reproductive success and thus natural selection does not affect these variations;
    - Eg. mutations that do not result in changes to proteins, and thus, no changes to phenotypes and these variations are maintained in the gene pool;
    - Eg. Mutations that result in changes in proteins can also be neutral eg. degeneracy of genetic code;

QWC: 1m for at least 3 different mechanisms.